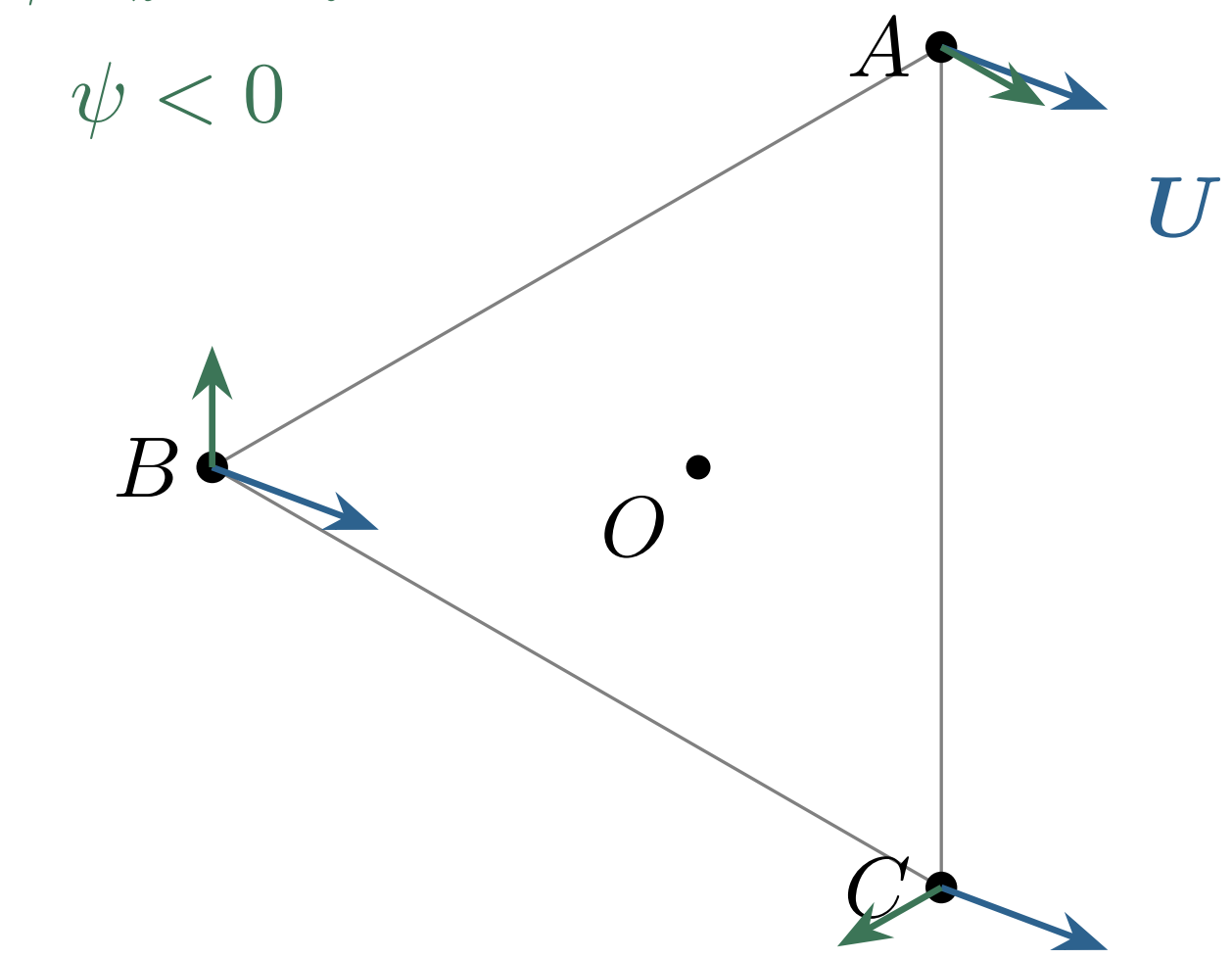

$$\boldsymbol{u}_i = \boldsymbol{U} + \psi \boldsymbol{e}_z \times \boldsymbol{r}_i$$

$$\boldsymbol{f}_i = -\kappa \boldsymbol{u}_i$$